

GPS: u-blox NEO-M8U

VCC Range: 2.7 – 3.6V

7-bit unshifted I2C address: 0x42

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[illegible]

The diagram illustrates the power supply section of a USB-to-serial module. It starts with a USB connector (labeled 'USB-C') providing 5V (VBUS), D+, D-, CC1, CC2, SHLD, and GND. The 5V line is connected to a 1.0uF capacitor (C2) and the input of a voltage regulator (U2, V_REG_AP2112K-3.3V). The regulator's output (3.3V) is connected to a 1.0uF capacitor (C4) and a 3.3V supply point. The regulator's enable pin (EN) is connected to the 5V line through a 10k resistor (R12). The regulator's ground pin (GND) is connected to the common ground. The D+ and D- lines are connected to a USB-to-serial converter chip (U1, MAX3232CPE) through 5.1k resistors (R13). The converter chip's ground is also connected to the common ground. The SHLD pin of the USB connector is connected to the shield of the USB cable.

LEDs and Jumpers

LEDs and Jumpers

Circuit diagram showing connections for LEDs and jumpers:

- I2C Section:**
 - 3.3V supply connected to the I2C bus.
 - Resistor R4 (1kOhm) is connected to the I2C bus.
 - Resistor R7 (2.2kOhm) is connected to the I2C bus.
 - I2C chip is connected to the bus.
 - SDA/CS and SCL/SCK lines are connected to the I2C bus.
 - Jumpers JP2 and JP3 are used to cut the I2C pull-up and pull-down respectively.
- LED Section:**
 - RED LED (D1) is connected to the I2C bus through a 1kOhm resistor.
 - BLUE LED (D2) is connected to the I2C bus through a 1kOhm resistor.
 - Jumpers JP4 and JP5 are used to close the jumper to enable SPI.

Labels in the diagram include: 3.3V, I2C, R4, R7, 1kOhm, 2.2kOhm, SDA/CS, SCL/SCK, D1, RED, D2, BLUE, JP2, JP3, JP4, JP5, GND, and various resistor values (1/10W-1%, 1/10W-1%).

LEDs and Jumpers

LEDs and Jumpers

3.3V

Cut jumper to remove I2C pullups from bus

I2C

R4 220Ω - 0603 - 1/10W - 1%

R7 220Ω - 0603 - 1/10W - 1%

SDA/CS

SCL/SCK

Cut jumpers to disable LEDs

D1 RED 1kΩ - 0603 - 1/10W - 1%

D2 BLUE 1kΩ - 0603 - 1/10W - 1%

3.3V

JP2

JP3

JP4

Close jumper to enable SPI

GND

[illegible]

LEDs and Jumpers

LEDs and Jumpers

Circuit diagram showing connections for LEDs and jumpers:

- I2C Section:**
 - 3.3V supply connected to the I2C pin.
 - Resistor R4 (1kOhm) and Resistor R7 (2.2kOhm) are connected to the I2C pin.
 - Resistor R6 (10kOhm) is connected to the I2C pin.
 - Jumper JP2 is located on the I2C pin.
 - Note: "Cut jumper to remove I2C pullups from bus"
- LED Section:**
 - Two LEDs are shown: RED1 and BLUE1.
 - RED1 is connected to the I2C pin.
 - BLUE1 is connected to the I2C pin.
 - Jumper JP3 is located on the I2C pin.
 - Note: "Close jumper to enable SPI"
- Other Connections:**
 - 3.3V supply connected to the D_SEL pin.
 - Jumper JP4 is located on the D_SEL pin.
 - Note: "Close jumper to enable SPI"

Headers

The diagram illustrates the pin connections for five headers: J1, J2, J3, J4, and J6. Headers J1, J2, J3, and J4 are Qwiic Right Angle headers, while J6 is a standard 6-pin header. All headers are connected to a 3.3V supply and GND. Header J6 is also connected to a 5V supply.

Header J1:

- Pin 1: GND
- Pin 2: VCC (3.3V)
- Pin 3: SDA
- Pin 4: SCL/SCK

Header J2:

- Pin 1: EXT_IN
- Pin 2: TP2/SA1
- Pin 3: RESET
- Pin 4: PPS

Header J3:

- Pin 1: GND
- Pin 2: VCC (3.3V)
- Pin 3: SDA
- Pin 4: SCL/SCK

Header J4:

- Pin 1: GND
- Pin 2: VCC (3.3V)
- Pin 3: SDA
- Pin 4: SCL/SCK

Header J6:

- Pin 1: GND
- Pin 2: SDA/CS
- Pin 3: RXI/MOSI
- Pin 4: TXD/MISO
- Pin 5: 5V
- Pin 6: SCL/SCK

Headers

The diagram illustrates the pin connections for five headers: J1, J2, J3, J4, and J6. Headers J1, J2, J3, and J4 are Qwiic Right Angle headers, while J6 is a standard 7-pin header. All headers are connected to a 3.3V supply and GND. Header J6 is also connected to a 5V supply.

Header J1:

- Pin 1: GND
- Pin 2: VCC
- Pin 3: SDA
- Pin 4: SCL/SCK

Header J2:

- Pin 1: EXT_IN
- Pin 2: TP2/SA1
- Pin 3: RESET
- Pin 4: PPS

Header J3:

- Pin 1: GND
- Pin 2: VCC
- Pin 3: SDA
- Pin 4: SCL/SCK

Header J4:

- Pin 1: GND
- Pin 2: VCC
- Pin 3: SDA
- Pin 4: SCL/SCK

Header J6:

- Pin 1: GND
- Pin 2: SDA/CS
- Pin 3: RXI/MOSI
- Pin 4: TXD/MISO
- Pin 5: 5V
- Pin 6: SCL/SCK
- Pin 7: 3.3V

Headers

The diagram illustrates the pin connections for five headers: J1, J2, J3, J4, and J6. Each header is represented by a red rectangle with pins numbered 1 to 4. The connections are as follows:

- J1:** Pin 1 to GND, Pin 2 to VCC (3.3V), Pin 3 to SDA/SCK, Pin 4 to SCL/SCK.
- J2:** Pin 1 to EXT_IN#, Pin 2 to TP2/SAT, Pin 3 to RESET, Pin 4 to PPS.
- J3:** Pin 1 to GND, Pin 2 to VCC (3.3V), Pin 3 to SDA/SCK, Pin 4 to SCL/SCK.
- J4:** Pin 1 to GND, Pin 2 to VCC (3.3V), Pin 3 to SDA/SCK, Pin 4 to SCL/SCK.
- J6:** Pin 1 to GND, Pin 2 to SDA/CS, Pin 3 to RXI/MOSI, Pin 4 to TXD/MISO, Pin 5 to 5V, Pin 6 to SCL/SCK, Pin 7 to 3.3V.

Hot Start Battery

Hot Start Battery

[illegible]

u.FL Antenna

J7

GPS_ANT1

FB1 FERRITE_BEAD-0603

VDD

R15

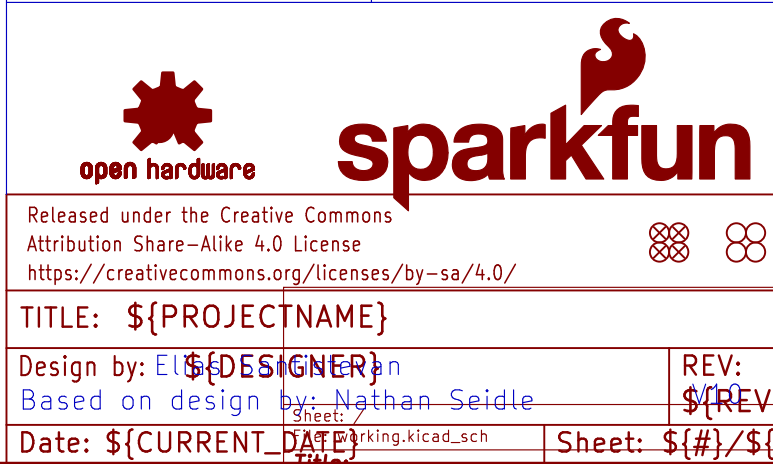
100OHM-0603-1/10W-1%

C5

47PF-0603-50V-5%

GND

GND



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TITLE: \${PROJECTNAME}		
Design by: Elliot Smith	Based on design by: Nathan Seidle	REV: V1.0
Date: 2016-07-27		Sheet: 1/1

Size: User	Date:	Rev:
KiCad E.D.A. eeschema 7.0.6		Id: 1/1

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